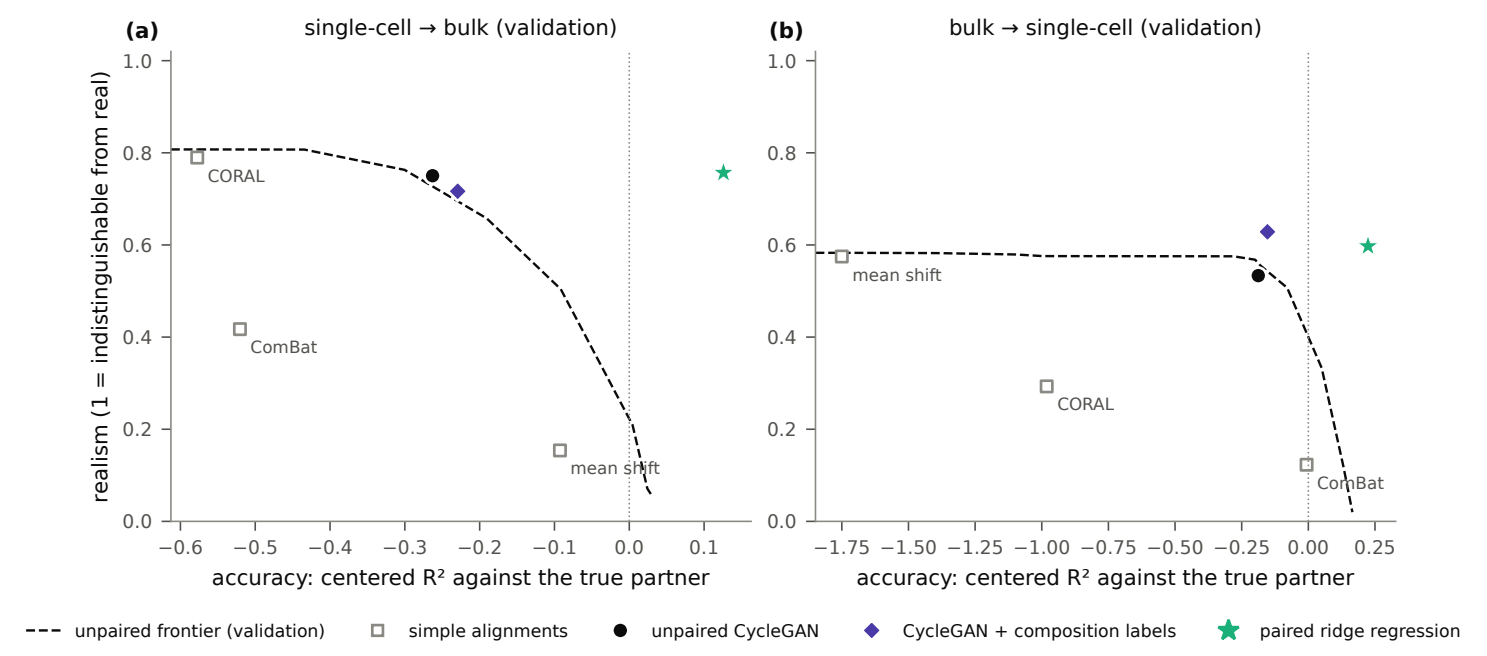
(a)
single-cell → bulk (validation)
realism (1 = indistinguishable from real)
1.0
0.8
0.6
0.4
0.2
0.0
CORAL
ComBat
mean shift
−0.6
−0.5
−0.4
−0.3
−0.2
−0.1
0.0
0.1
accuracy: centered R² against the true partner
(b)
bulk → single-cell (validation)
mean shift
CORAL
ComBat
−1.75
−1.50
−1.25
−1.00
−0.75
−0.50
−0.25
0.00
0.25
accuracy: centered R² against the true partner
unpaired frontier (validation)
simple alignments
unpaired CycleGAN
CycleGAN + composition labels
paired ridge regression